

# NGS-DOSE on the 1000 Genomes 30× cohort: a trio-focused assessment

as of 2026-09-29; results from the tables python -m report writes, method text fixed

## The question

Can the copy number of a sequence that a reference genome collapses — the ribosomal DNA arrays, several hundred copies per genome — be measured from ordinary short-read whole-genome sequencing? An assay of rDNA copy number (ddPCR) exists for only a dozen of these lines, so the answer has to come mostly from things that are known: sequence of known copy number in every genome, Mendelian transmission in trios, the same genome counted two ways, long-read assemblies of the same people for the satellite arrays that the same machinery measures, and the assay where it exists.

## What is measured

Reads are assigned to a class (45S rDNA, 5S rDNA, the distal junction, ten satellite families, the telomeric repeat) by 31-mers: for 45S, 5S and the distal junction, k-mers found nowhere else in GRCh38 or CHM13; for the satellite families, k-mers found in no other family and nowhere in CHM13 outside annotated satellite; for the telomeric repeat, the six canonical (TTAGGG)<sub>n</sub> 31-mers, unfiltered. Where the aligner put a read is not used. Fragment 5' ends are counted on the unit and in 800 single-copy control regions. A per-library Poisson spline of fragment-end density on fragment GC content, fitted on the controls, gives the expected count of any sequence; copy number is  $2 \times \text{observed} / \text{expected}$  per 250-bp window, and the windows of the 45S unit are calibrated across the cohort with the scale set by anchor windows on which three Illumina chemistries agreed. Two counting modes: scan reads the whole CRAM, fetch retrieves only the controls and the intervals where the aligner places class reads (learned from whole-file scans of this pipeline's alignments; a pipeline aligned otherwise needs its own). Cohort: the 1000 Genomes 30× CRAMs (NYGC, NovaSeq 2×150, GRCh38), all lymphoblastoid cell lines. Three 45S estimators travel through every table, two of them NGS-DOSE's. '45S, NGS-DOSE calibrated': the cohort model  $\log C(i,w) = c(i) + a(w) + e(i,w)$  over every retained 250-bp window  $w$  of the unit, fitted by median polish across samples  $i$  with the window efficiencies  $a(w)$  pinned to a median of zero over the anchor windows; the estimate is  $\exp(c(i))$ . '45S, NGS-DOSE single-sample':  $2 \times \text{observed} / \text{expected}$  fragment ends over the anchor windows alone, under the sample's own GC model, with no information from any other sample. '45S, 18S depth ratio (published)':  $2 \times \text{fragment ends in the 18S gene} / (\text{positions} \times \text{the control regions' mean rate})$ , no GC model, no calibration — the estimator of published studies, computed from the same reads for comparison; it is not NGS-DOSE's estimate.

## How it is tested

In a trio, a child's copy number is the mean of the parents' plus segregation; measurement error is not inherited. Values are centred within population; the correlations are Pearson's. The child-midparent correlation cannot reach 1 even for a perfectly measured heritable trait, because half of a child's variance is segregation, which the midparent does not predict ( $r$  is bounded by about  $\sqrt{(1 + \rho)/2}$ , 0.71 at  $\rho = 0$ ). The least-squares slope  $b$  of child on midparent has no such ceiling, so the slope, corrected for the spousal correlation  $\rho$ , is the statistic that answers what share of the measured variance is real: reliability  $R = b - \rho(1 - b)$ , 1 for a perfectly measured heritable trait and 0 for pure error, with family-bootstrap 95% intervals and a one-sided  $p$ -value for the slope from 1,000 permutations of children among families. The test is applied to every metric in four groups. The rDNA estimators are the claim. The satellite arrays are positive controls: their mass is a property of the genome, measured by the same k-mer machinery, and must be inherited. Known copy numbers (held-out autosomal sequence, the distal junction) have nothing to inherit. The culture's and the library's properties (mitochondrial and EBV content, depth, duplicate rate, GC bias, insert size) are not in the nuclear genome. A method that measured library artefacts would light up the last group; one that measured nothing would light up none. The same test is then run on the fetch counts alone, and the satellite masses are held against HPRC assemblies of the same people.

## Results

3,202 of 3,202 genomes, 602 complete trios, as of 2026-09-29. Known copy numbers: held-out autosomal sequence  $1.997 \pm 0.008$  (expected 2); the distal junction  $9.99 \pm 0.33$  (expected 10); sex from the reads agrees with the pedigree in 3,200 of 3,202. Transmission, 45S calibrated estimate: reliability 0.95 (0.86–1.04), child-midparent  $r$  0.69, spousal  $r$  0.06. 5S: 1.00 (0.88–1.15). Positive controls, 9 satellite families: reliability 0.44 to 1.00 (HSat1B, on Yq, excluded: father-to-son only); the telomeric repeat, which changes with age and in culture, 0.35. Negative controls: culture and library -0.31 to 0.15; known copy numbers -0.04 to 0.72. Fetch counts alone: 45S reliability 0.95; fetch/scan agreement of the 45S estimate across genomes  $r$  1.0000. Targeted fetch against whole-file scan, 45S: median 0.9996 (0.9988–0.9999) over 3,202 genomes. Parents and children were sequenced apart: 597 of 602 children in release batch 698, 1097 of 1204 parents in 2,504. In 517 of 602 families the child was sequenced in a different batch from both parents (85 share one), and a batch that reads a generation on another scale moves  $R$  with it: 5S, NGS-DOSE calibrated spreads  $1.08\times$  as much in the children as in their parents and reads  $R$  1.02, or 0.95 with the children on their parents' scale; HSat2 array spreads  $1.08\times$  as much in the children as in their parents and reads  $R$  1.03, or 0.96 with the children on their parents' scale. Against ddPCR (Potapova et al. 2025) on 12 lymphoblastoid lines: NGS-DOSE  $0.96\times$  the assay,  $r$  0.94; CONKORD  $0.99\times$ ,  $r$  0.87; the 18S depth ratio  $1.09\times$ ,  $r$  0.93. Assemblies: 200 of these genomes have an HPRC release-2 assembly. For 8 of 10 satellite families a person's estimate and their assembly agree to within a robust SD of 2–8% (HSat1A, HSat1B, HSat3, bSat, aSatHOR, ACRO, CER, SATR); across people the estimates track the assemblies at  $r$  0.76–1.00 where people differ by 10% or more (HSat1A, HSat1B, HSat3, bSat, ACRO), and  $r$  is bounded by how little people differ where the assembly's between-person CV is a few percent.

## Limits

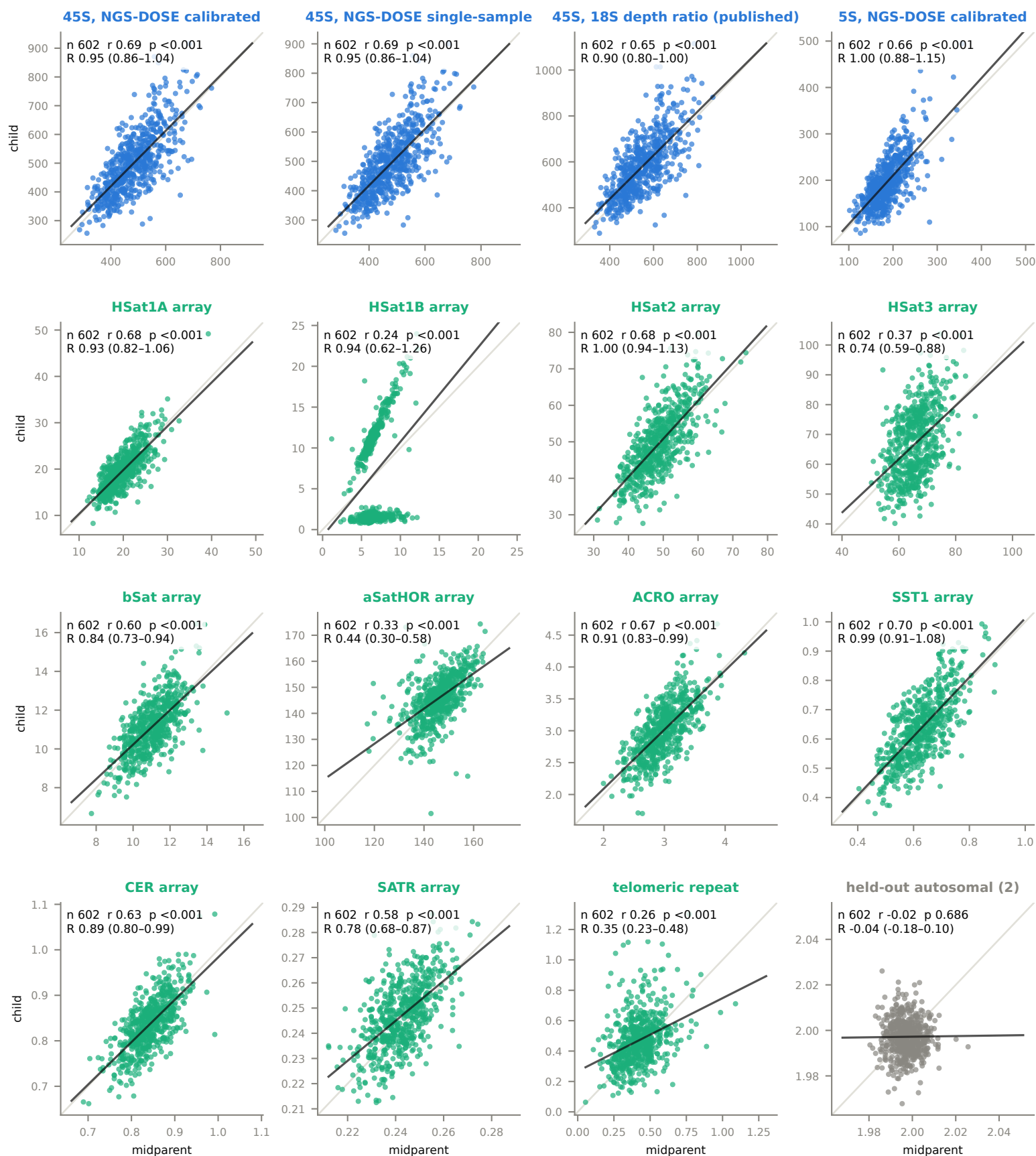
What this does not show: an absolute calibration of rDNA copy number beyond the dozen lines with a ddPCR value (elsewhere the scale rests on unit windows on which three chemistries agree); anything about DNA that is not from a lymphoblastoid cell line, a chemistry other than NovaSeq 2×150, or an alignment pipeline other than NYGC's, on which the targeted fetch's sinks were learned; and, because people differ in 45S copy number far more than any competent estimator errs, the trios show that the measured variation is inherited, not which rDNA estimator is best — that ranking rests on the same people sequenced on two technologies (the pilot), where the calibrated estimate reproduced and the depth ratio did not.

## Transmission of every metric, at a glance

|                                                                                  |                                  | child-father r | child-mother r | child-midparent r | spousal r | midparent slope | reliability R | R from fetch |
|----------------------------------------------------------------------------------|----------------------------------|----------------|----------------|-------------------|-----------|-----------------|---------------|--------------|
| rDNA copy number<br>(the claim)                                                  | 45S, NGS-DOSE calibrated         | 0.50           | 0.50           | 0.69              | 0.06      | 0.95            | 0.95          | 0.95         |
|                                                                                  | 45S, NGS-DOSE single-sample      | 0.50           | 0.50           | 0.69              | 0.06      | 0.95            | 0.95          | 0.95         |
|                                                                                  | 45S, 18S depth ratio (published) | 0.48           | 0.47           | 0.65              | 0.06      | 0.91            | 0.90          | 0.90         |
|                                                                                  | 5S, NGS-DOSE calibrated          | 0.49           | 0.44           | 0.66              | -0.02     | 1.02            | 1.02          | 1.02         |
| satellite arrays,<br>mass (genomic:<br>positive controls)                        | HSat1A array                     | 0.48           | 0.50           | 0.68              | 0.05      | 0.94            | 0.93          |              |
|                                                                                  | HSat1B array                     | 0.24           | 0.01           | 0.24              | -0.13     | 0.93            | 0.94          |              |
|                                                                                  | HSat2 array                      | 0.48           | 0.48           | 0.68              | 0.01      | 1.03            | 1.03          |              |
|                                                                                  | HSat3 array                      | 0.29           | 0.23           | 0.37              | 0.01      | 0.74            | 0.74          |              |
|                                                                                  | bSat array                       | 0.41           | 0.44           | 0.60              | -0.01     | 0.84            | 0.84          |              |
|                                                                                  | aSatHOR array                    | 0.22           | 0.25           | 0.33              | 0.00      | 0.44            | 0.44          |              |
|                                                                                  | ACRO array                       | 0.47           | 0.49           | 0.67              | 0.02      | 0.91            | 0.91          |              |
|                                                                                  | SST1 array                       | 0.52           | 0.46           | 0.70              | -0.01     | 0.99            | 0.99          |              |
|                                                                                  | CER array                        | 0.44           | 0.47           | 0.63              | 0.05      | 0.89            | 0.89          |              |
|                                                                                  | SATR array                       | 0.42           | 0.42           | 0.58              | 0.05      | 0.79            | 0.78          |              |
| known copy number<br>(nothing to inherit<br>except the DJ's<br>whole-copy steps) | telomeric repeat                 | 0.17           | 0.19           | 0.26              | -0.00     | 0.35            | 0.35          | 0.28         |
|                                                                                  | held-out autosomal (2)           | 0.00           | -0.03          | -0.02             | 0.01      | -0.03           | -0.04         | -0.04        |
|                                                                                  | distal junction (10)             | 0.33           | 0.45           | 0.54              | 0.01      | 0.73            | 0.72          | 0.72         |
| culture and library<br>(not in the nuclear<br>genome)                            | mitochondrial genomes per cell   | 0.08           | 0.11           | 0.13              | 0.10      | 0.15            | 0.07          | 0.07         |
|                                                                                  | EBV episomes per cell            | 0.07           | 0.18           | 0.16              | 0.16      | 0.26            | 0.15          | 0.15         |
|                                                                                  | sequencing depth                 | -0.04          | -0.04          | -0.06             | 0.02      | -0.05           | -0.07         | -0.07        |
|                                                                                  | duplicate-flagged fraction       | 0.06           | 0.05           | 0.07              | 0.18      | 0.13            | -0.03         | -0.03        |
|                                                                                  | library GC bias                  | 0.02           | -0.01          | 0.01              | 0.10      | 0.01            | -0.09         | -0.09        |
|                                                                                  | insert size                      | -0.08          | -0.07          | -0.10             | 0.20      | -0.09           | -0.31         | -0.31        |

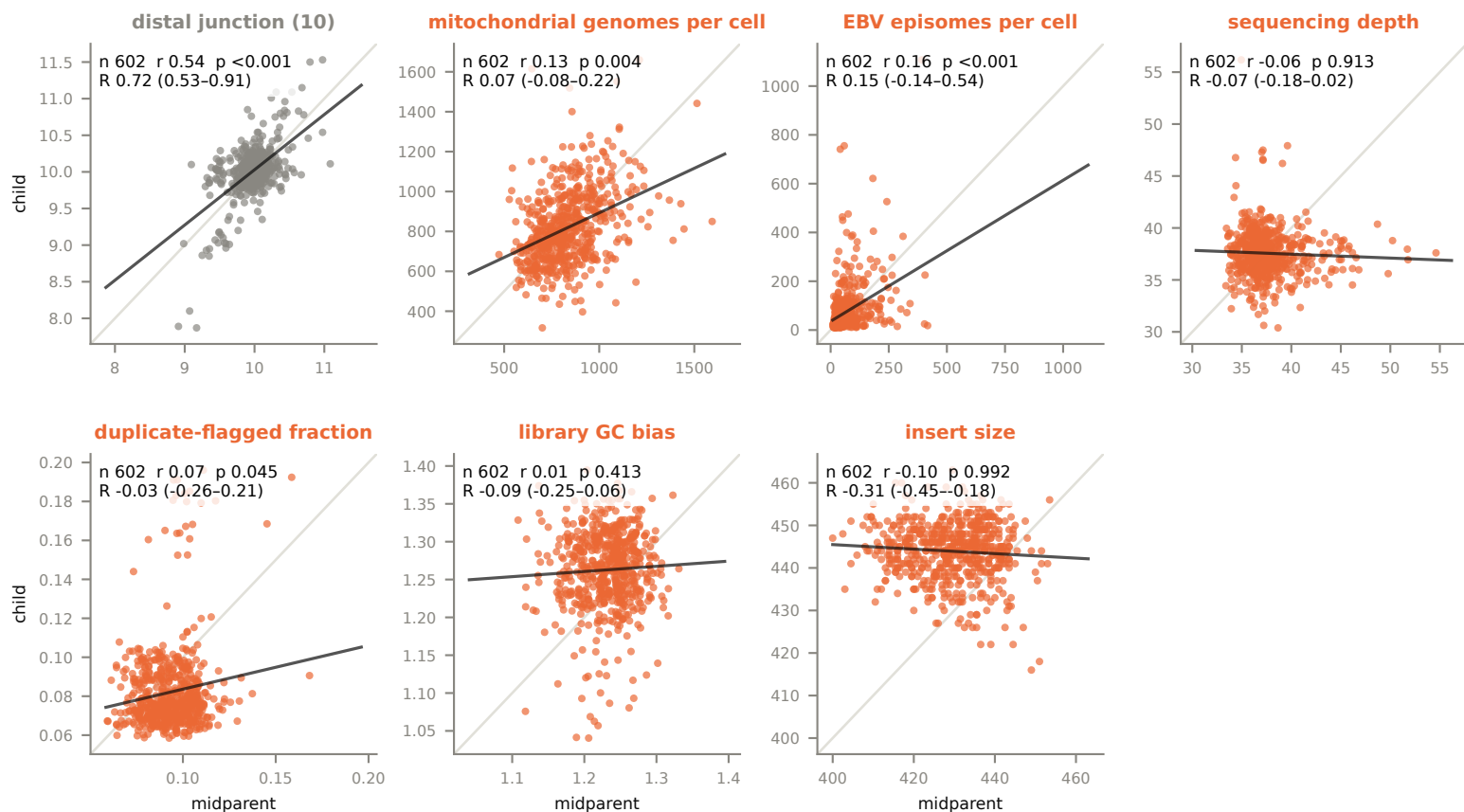
Colour and number: the statistic, from 0 (white) to 1 (blue); values are within population, over the complete trios. A metric that is inherited reads near 1 in the correlation columns and in R; a property of the culture or the library reads near 0; sequence of known copy number has no variation to inherit and reads near 0 too. HSat1B lives mostly on Yq and passes from father to son only, which dilutes its midparent statistics by design. The last column repeats R with the cohort layer and the trio test run on the fetch counts alone.

# Child against midparent, every metric



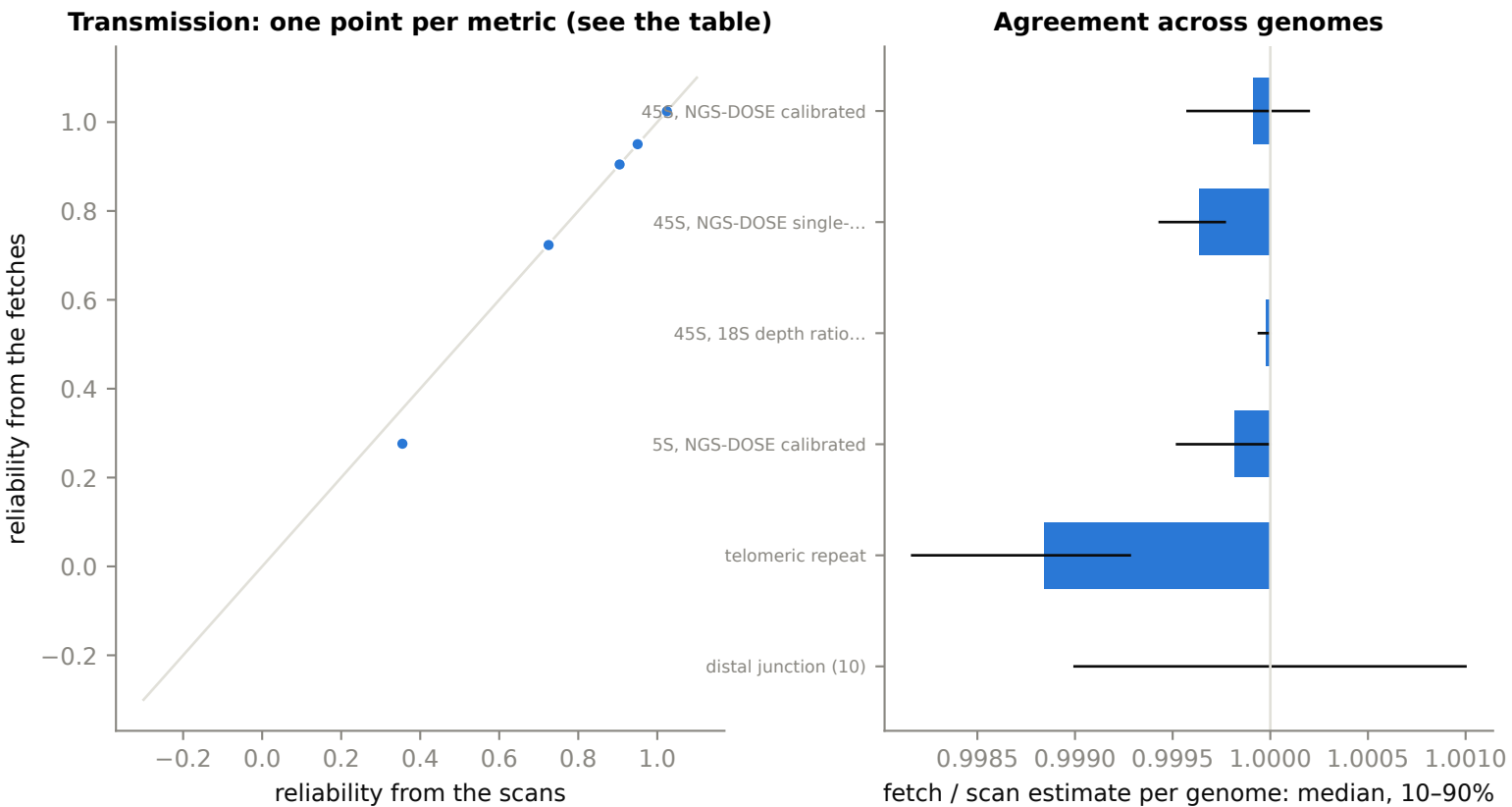
Raw values (not centred); grey diagonal: child equals midparent; black line: least squares. Printed: the report's statistics, computed within population (n trios, Pearson child-midparent r, the permutation p of the midparent slope, reliability R with its family-bootstrap 95% interval). Colours: blue rDNA estimators, green satellite arrays (positive controls), grey known copy numbers, orange culture and library (negative controls).

## Child against midparent, every metric (continued)



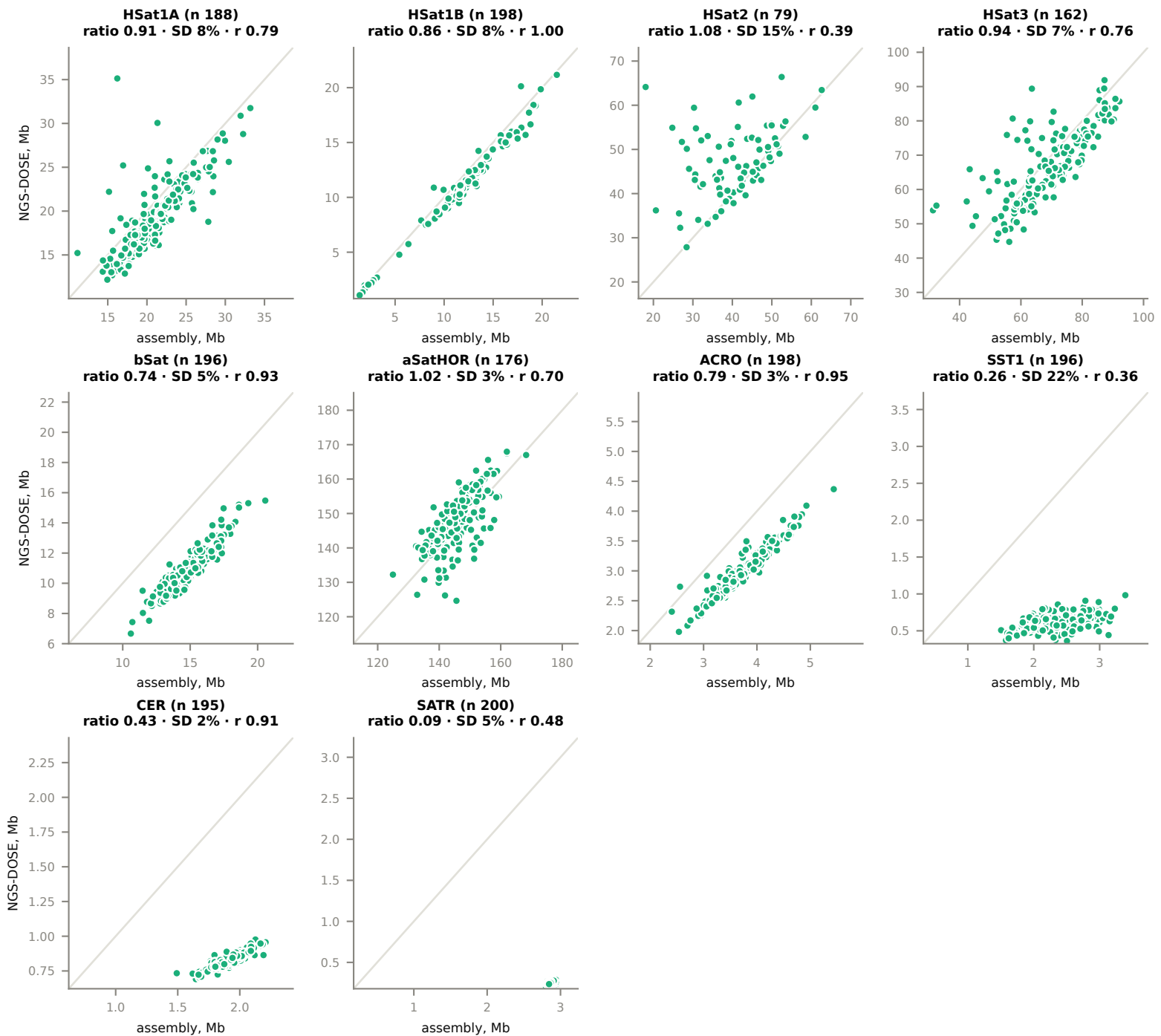
Raw values (not centred); grey diagonal: child equals midparent; black line: least squares. Printed: the report's statistics, computed within population (n trios, Pearson child-midparent r, the permutation p of the midparent slope, reliability R with its family-bootstrap 95% interval). Colours: blue rDNA estimators, green satellite arrays (positive controls), grey known copy numbers, orange culture and library (negative controls).

# The same test on the fetch counts alone



The cohort layer (window calibration, control-region PCs) and the trio test were run on the fetch counts of 3,202 genomes without reference to the scans. Columns made from the same reads in both modes come out identical and are omitted (held-out autosomal (2), mitochondrial genomes per cell, EBV episomes per cell, sequencing depth, duplicate-flagged fraction, library GC bias, insert size, chrX, chrY). The classes differ by what the sinks miss and by a calibration learned twice.

# Satellite arrays against long-read assemblies of the same people



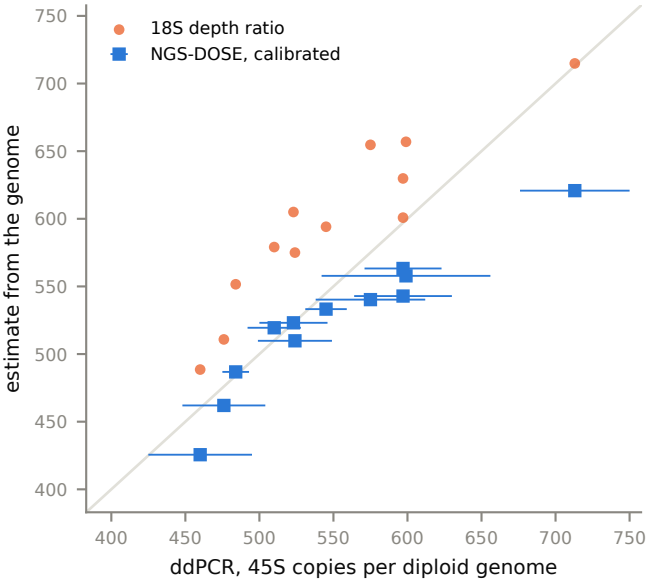
200 of the genomes have HPRC release-2 assemblies; the CenSat annotation of both haplotypes gives the size of every satellite array, the same kind of sequence the k-mer machinery measures. A genome is left out of a class when the gaps its annotation marks (a GAP record labelled with the family, or a standalone GAP next to the array) amount to more than 2% of the class; 100-bp placeholder gaps of unknown size are counted but leave no one out (people compared with one: HSat1A 15 of 188, HSat1B 2 of 198, HSat2 2 of 79, HSat3 15 of 162, bSat 9 of 196, aSatHOR 152 of 176, ACRO 4 of 198, SST1 6 of 196, SATR 2 of 200). Grey diagonal: equality. The relative measures ( $\beta$ -satellite, CER, ACRO, SST1, SATR) sit below it by a constant factor, their k-mer recall. Each title gives the median estimate-to-assembly ratio, the robust SD of the log ratio (how far one person's two values disagree) and the correlation across people, which is bounded by how little people differ: where the assembly's between-person CV is a few percent ( $\alpha$ -satellite HORs, SATR)  $r$  is low although each person agrees to within that SD. Assemblies collapse the rDNA itself and are no truth for it.

Sequence of known copy number, fetch against scan, and ddPCR

| what                                            | expected | measured                 | n     |
|-------------------------------------------------|----------|--------------------------|-------|
| held-out autosomal sequence (80 regions)        | 2        | 1.997 ± 0.008            | 3,202 |
| chrX in men with one X (60 regions)             | 1        | 0.992 ± 0.006            | 1,596 |
| chrX / chrY, HG01683 (a man with two X and a Y) | 2 / 1    | 1.97 / 0.97              | 1     |
| chrX in women with an intact culture            | 2        | 1.939 ± 0.038            | 1,543 |
| chrY in men with an intact Y (40 regions)       | 1        | 0.983 ± 0.044            | 1,585 |
| chrY in women                                   | 0        | at most 0.0124           | 1,605 |
| distal junction (one per acrocentric short arm) | 10       | 9.99 ± 0.33              | 3,202 |
| sex inferred from the reads                     | pedigree | agrees in 3,200 of 3,202 |       |
| fetch / scan, 45S                               | 1        | 0.9996 (0.9988–0.9999)   | 3,202 |
| fetch / scan, 5S                                | 1        | 0.9999 (0.9992–1.0000)   | 3,202 |
| fetch / scan, distal junction                   | 1        | 0.9977 (0.9965–0.9986)   | 3,202 |
| fetch / scan, held-out autosomal                | 1        | 1.0000 (1.0000–1.0000)   | 3,202 |
| fetch / scan, chrM                              | 1        | 1.0000 (1.0000–1.0000)   | 3,202 |
| fetch / scan, telomeric repeat, mass            | 1        | 0.9988 (0.9951–1.0003)   | 1,454 |

Every genome carries sequence whose copy number is not in question, measured under the same fragment-GC model as the rDNA: held-out autosomal regions and the X and Y, counted by their alignment position, and the distal junction, a 400-kb sequence present once on each of the ten acrocentric short arms — multi-copy, paralogous and acrocentric, the kind of sequence the rDNA is — counted by the same k-mer path as the rDNA. Women read the X, and the distal junction reads, a few percent below expectation on average. Both are late-replicating sequence, but in women the two deficits show no detectable correlation ( $r = -0.00, -0.05$  to  $0.05$ ), which weakens a shared S-phase explanation without ruling out a small one. The junction's scale is pinned to the cohort's mode: on the fragment-GC model's scale its core reads 9.78. By the whole numbers of copies called along the unit, relative to ten complete copies: 7 genomes at  $-2$ , 114 at  $-1$ , 3,011 at  $0$ , 32 at  $+1$ , 7 at  $+2$  and 2 at  $+3$ ; 220 genomes carry a copy that holds only part of the unit (the level of the others has a robust SD of 0.13 copies). A step is a structural variant of an acrocentric short arm; where a carrier parent and a child were both counted it was transmitted in 15 of 47 pairs, of which 1 fits neither transmission nor its absence and is left unclassified, and 1 child(ren) carry a step neither counted parent has.

Against ddPCR (Potapova et al. 2025)



| estimate                 | lines | ÷ ddPCR | diff  % | r     | Spearman |
|--------------------------|-------|---------|---------|-------|----------|
| NGS-DOSE, calibrated 45S | 12    | 0.957   | 4.9     | 0.945 | 0.972    |
| CONKORD (their reads)    | 12    | 0.994   | 4.0     | 0.874 | 0.965    |
| 18S depth ratio          | 12    | 1.093   | 8.8     | 0.926 | 0.916    |

12 lymphoblastoid lines with a NovaSeq genome, 9 counted in this run; the rest fetched from the same CRAMs and calibrated with the cohort's saved efficiencies, or counted from a second NovaSeq pipeline. Assay replicate CV 5.0%. The first external check of the absolute level.